

# 1 Transaction Processing

What are the primary considerations regarding transaction processing?

You must follow the ACID acronym:

**Atomicity:** A transaction must either completely succeed or completely fail. This means that in the case that a section of the transaction fails, all changes should be rolled back.

**Consistency:** The state before and after should be the same. This could mean that the total money in the database is the same for a bank or there is only one person with an item in a game.

**Isolation:** If a transaction is processing, nothing else can transact with the data being operated on.

**Durability:** If a transaction has occurred, then it should be saved on permanent storage so it is committed even in the event of system failure (ram data lost)

## 1.1 Questions

### 1.1.1 1(a):

**RestaurantReview** is a website that allows users to leave reviews and ratings for different restaurants.

The website uses a database with the following structure.

The database management system ensures referential integrity is maintained.

Whenever a review is added to the system, the restaurant's average rating is updated. This transaction is **ACID**.

The **A** in **ACID** refers to **Atomic**.

Describe what is meant by the term 'Atomic' in the context of **ACID** transactions. You should refer to the example of a review being added.

ANSWER

Atomic means that a transaction either occurs completely or doesn't occur at all. In the context of a review, a transaction would include creating the review and updating the restaurants average rating. If one fails the other doesn't occur.

### 1.1.2 1.b

Whenever a review is added to the system, the restaurant's average rating is updated. This transaction is **ACID**.

The **A** in **ACID** refers to **Atomic**.

State what the letters **CID** refer to in **ACID**.

ANSWER

C refers to consistency, I refers to isolation and D refers to durability.